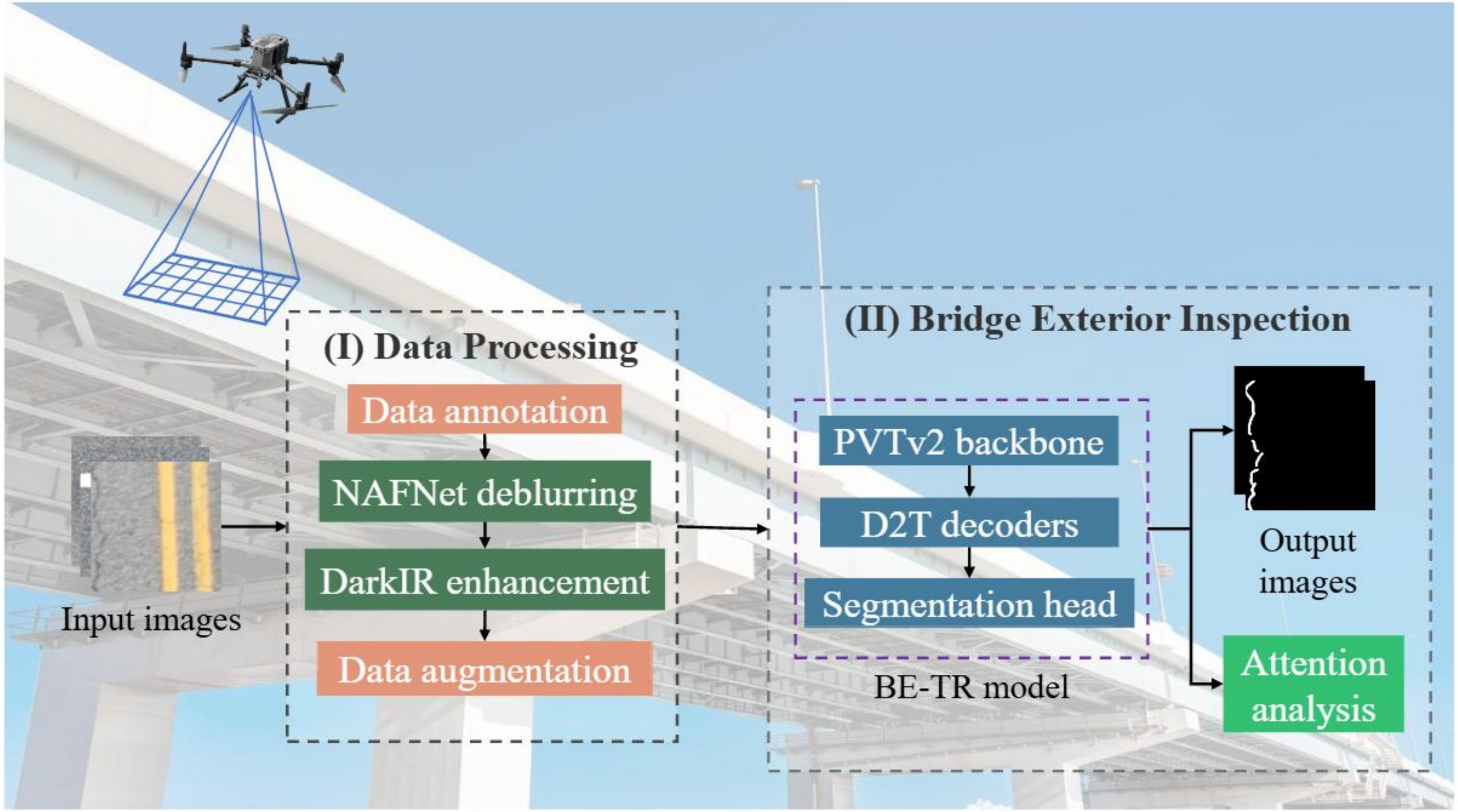
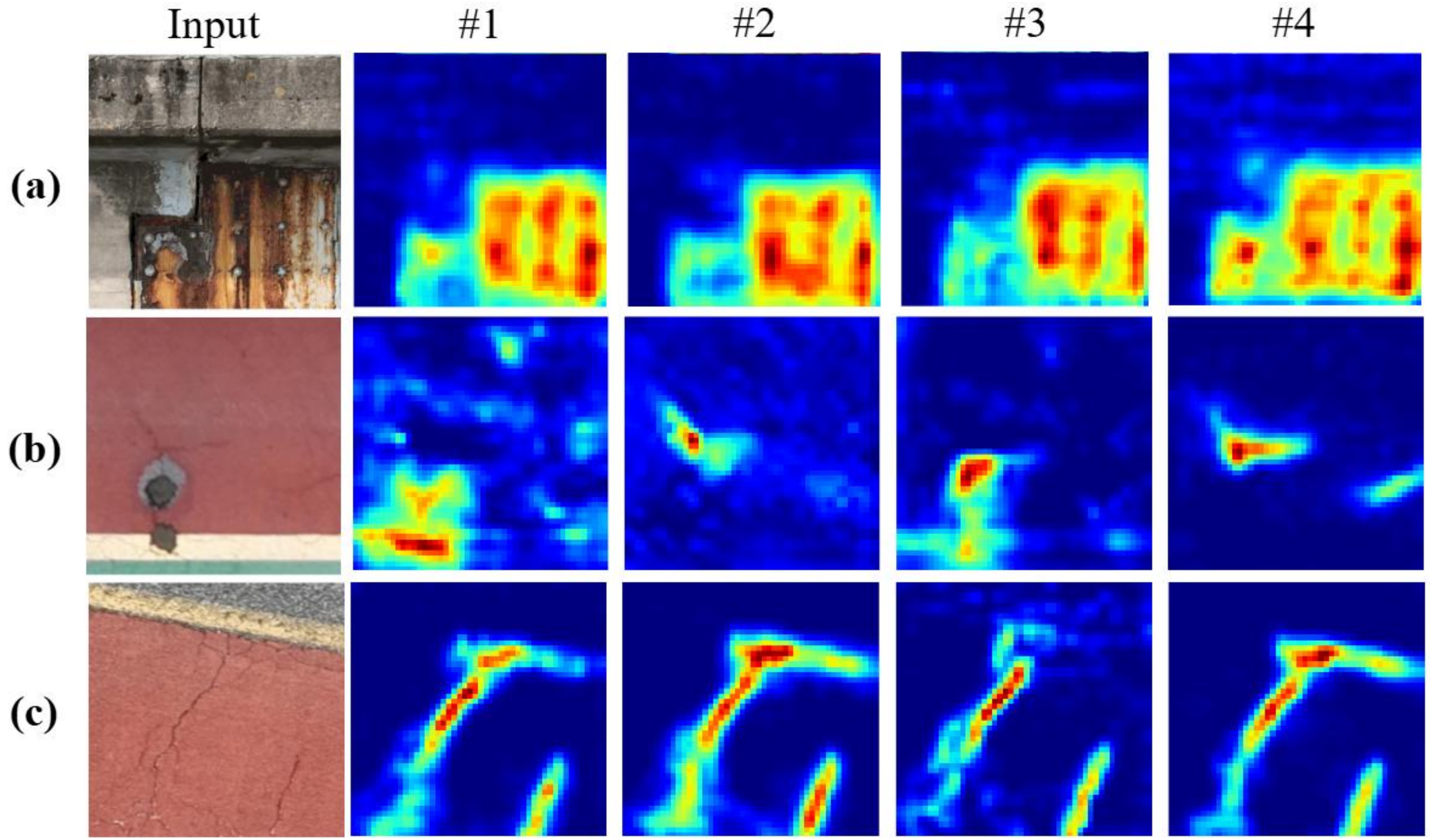
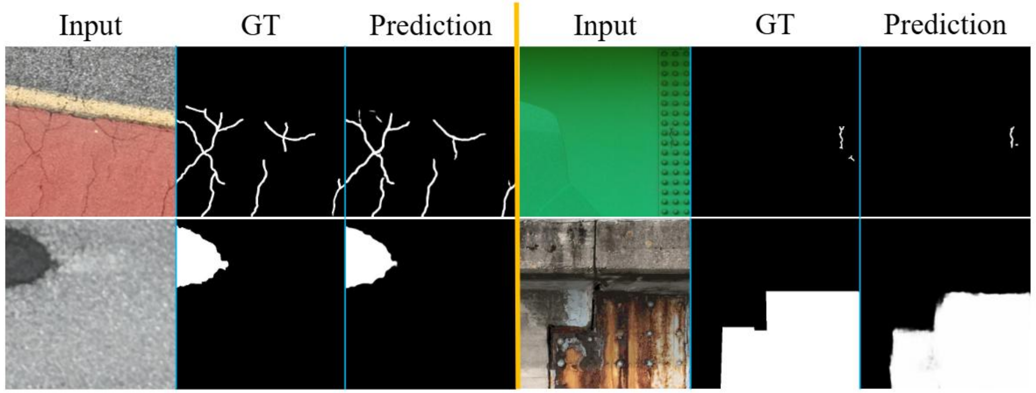
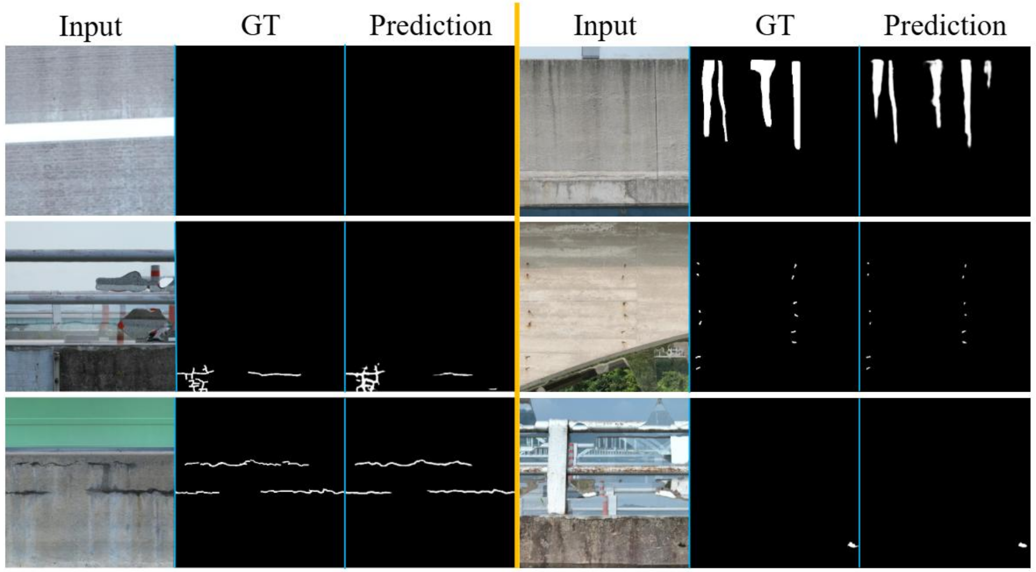

(I) Data Processing
Data annotation
NAFNet deblurring
DarkIR enhancement
Data augmentation
Input images
(II) Bridge Exterior Inspection
PVTv2 backbone
D2T decoders
Segmentation head
BE-TR model
Output images
Attention analysis



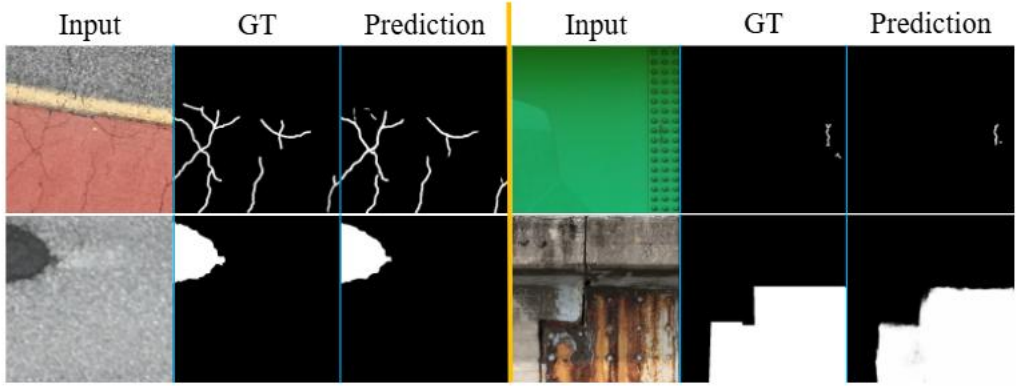
Input
#1
#2
#3
#4
(a)
(b)
(c)



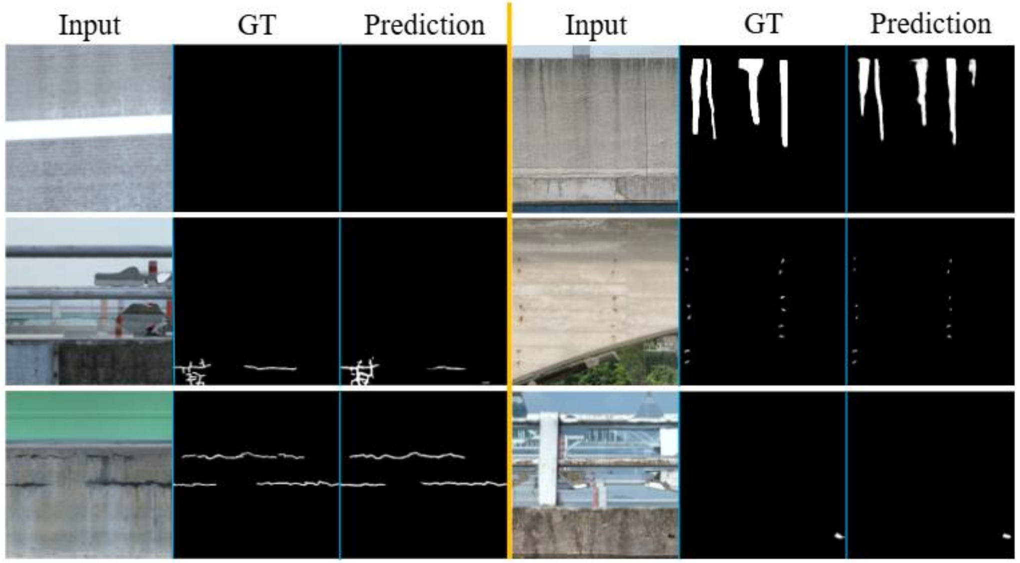
(a) Predicted mask results on asphalt and steel bridges, including asphalt cracks, pothole, steel corrosion, and paint peeling.



(b) Predicted mask results on concrete bridges, including concrete crack, spalling, efflorescence, leak, and exposed rebar.



(a) Predicted mask results on asphalt and steel bridges, including asphalt cracks, pothole, steel corrosion, and paint peeling.



(b) Predicted mask results on concrete bridges, including concrete crack, spalling, efflorescence, leak, and exposed rebar.

Figure 9: Qualitative examples of SDD generated by the BE-TR model across various bridge types are presented. For each sample, the first column displays the input image, the second column presents the ground truth (GT) defect mask, and the third column illustrates the predicted mask.

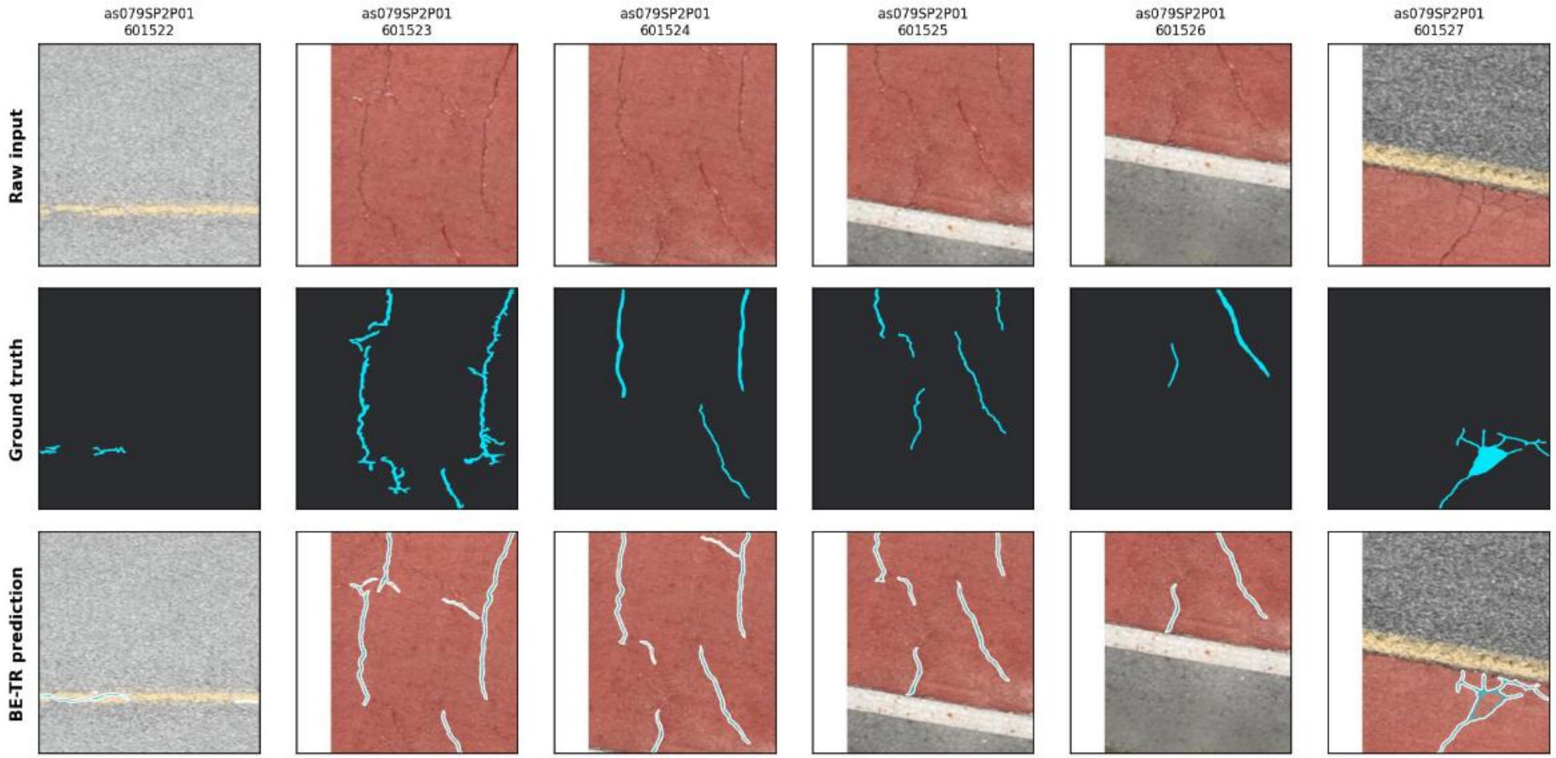
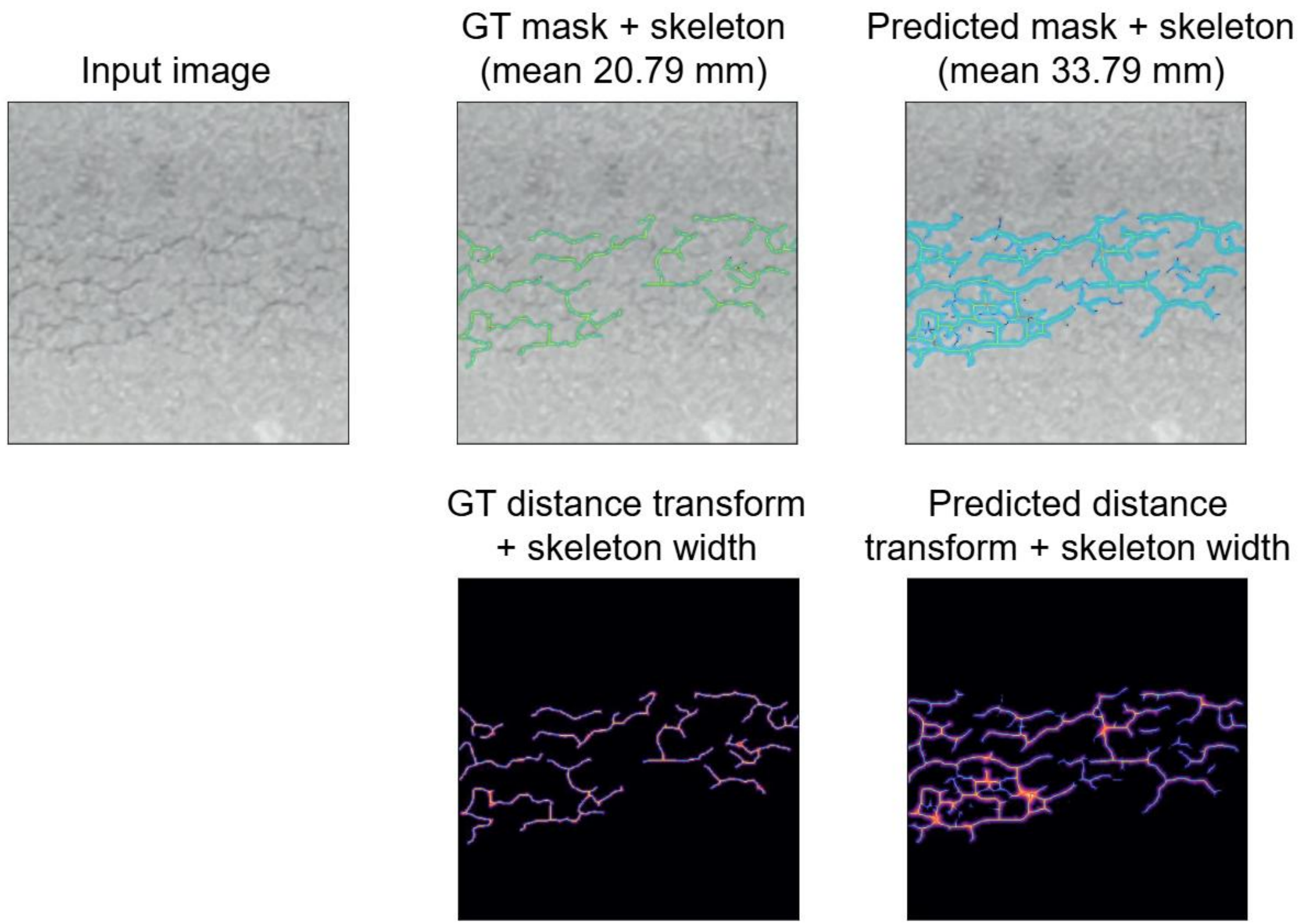


Figure 11: Six sequential frames from a bridge inspection sequence are detected using the BE-TR model. The first column shows the input images, and columns two through six display the corresponding overlay results.


Input image
GT mask + skeleton
(mean 20.79 mm)
Predicted mask + skeleton
(mean 33.79 mm)
GT distance transform
+ skeleton width
Predicted distance
transform + skeleton width

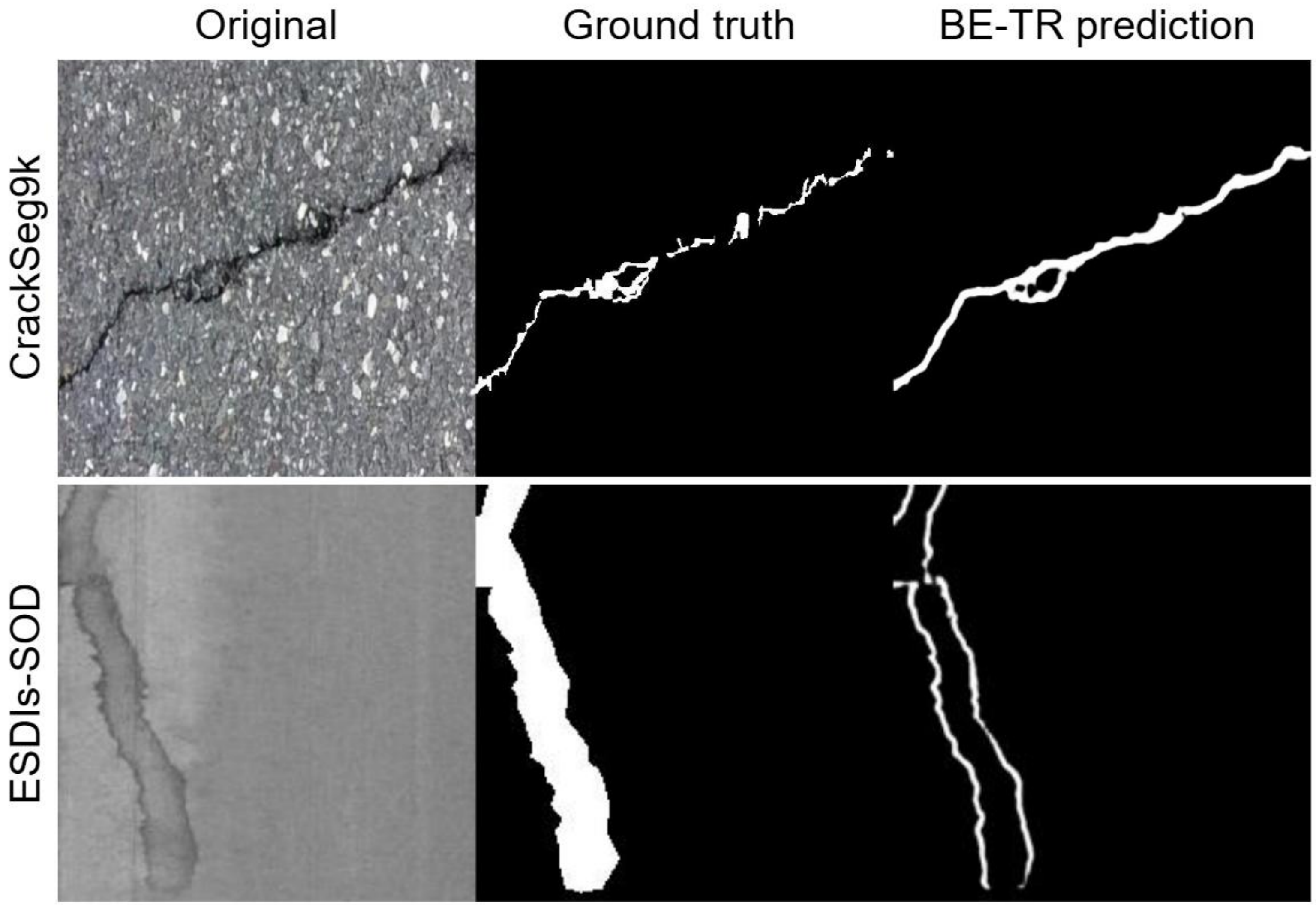


Figure 13: BE-TR model predictions on additional datasets, including CrackSeg9k and ESDIs-SOD.